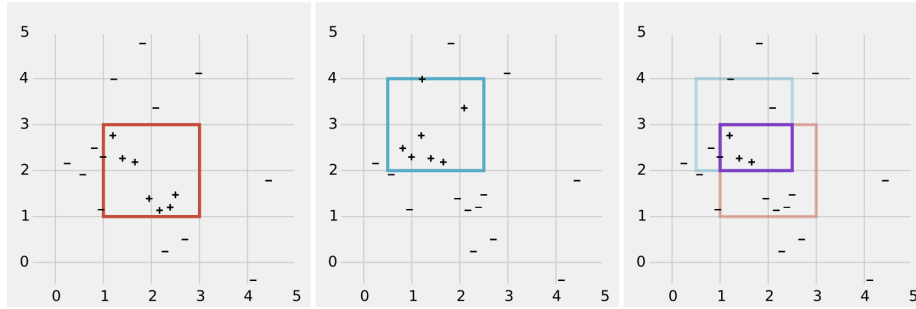


1 VC Dimension

Recall that we call a set of points *shattered* by a class of functions H if all possible $\{-1, +1\}$ labelings of the points can be produced by some function in H . The *Vapnik-Chervonenkis* (VC) dimension is the size of the largest set of points that can be shattered by the hypothesis space.

In this problem, we will explore the hypothesis space where each hypothesis is a combination of two simpler hypotheses. More precisely, given two hypotheses h_1 and h_2 , we define $h = h_1 \cap h_2$ as a new hypothesis that labels an example $+1$ only if both h_1 and h_2 give the label $+1$, otherwise, it is labeled -1 . We can extend this to *sets* of hypotheses: given two sets of hypotheses H_1 and H_2 , define $H^* = \{h_1 \cap h_2 : h_1 \in H_1, h_2 \in H_2\}$ as the set of all intersections of hypothesis pairs from the two classes H_1 and H_2 .

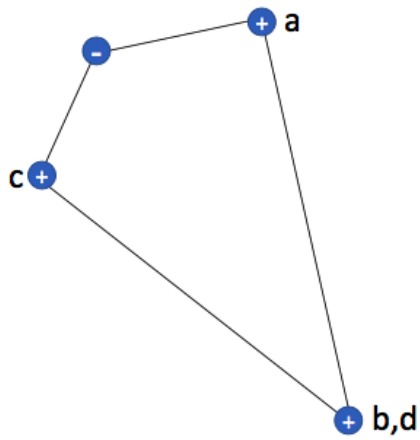
As an example, let H_1 be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is larger than some threshold a . Let H_2 be the set of classifiers in $\mathbb{R}$ that assigns the label $+1$ if the example is smaller than some threshold b . Then H^* would be the set of all intervals (a, b) in $\mathbb{R}$ that assigns $+1$ if the example is inside the interval. Another example is when H_1 and H_2 is the set of all (axis-aligned) squares in $\mathbb{R}^2$, H^* is the set of all axis-aligned rectangles. This example is illustrated below. On the left, we have a single square classifier h_1 ; in the middle we again have a square classifier h_2 ; and on the right, we have $h_1 \cap h_2$, which is a rectangle classifier.



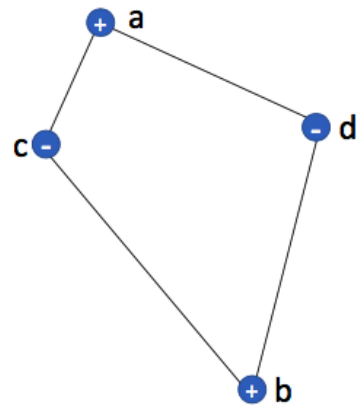
Keep in mind that these are only examples. We are looking for results that can apply generally to any pair of hypotheses classes.

1. Suppose that the *shattering coefficient* of H_1 is $H_1[n]$ (i.e. the maximum number of ways that the hypothesis space H_1 can label a set of n points is $H_1[n]$). Similarly, suppose that the shattering coefficient of H_2 is $H_2[n]$. Show that $H^*[n] \leq H_1[n]H_2[n]$.

2. For each one of the following function classes, find the VC dimension. State your reasoning.
 - i. Half spaces in $\mathbb{R}$, where examples on one side of the boundary are labeled $+1$, and examples on the other side are labeled -1 .



Case 1: the labeling above is not achievable



Case 2: the labeling above is not achievable

ii. Half spaces in $\mathbb{R}^2$, where examples on one side of the line are labeled $+1$, and examples on the other side are labeled -1 .

iii. Axis-aligned squares in $\mathbb{R}^2$, where points are labeled $+1$ inside the square, and -1 outside (as in the illustrations above).

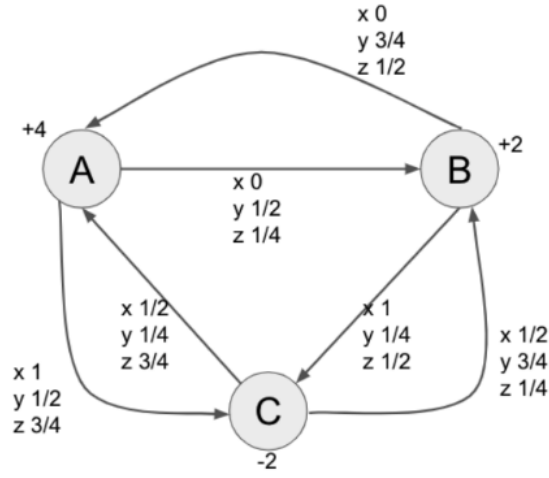


Figure 1: MDP

2 Reinforcement Learning

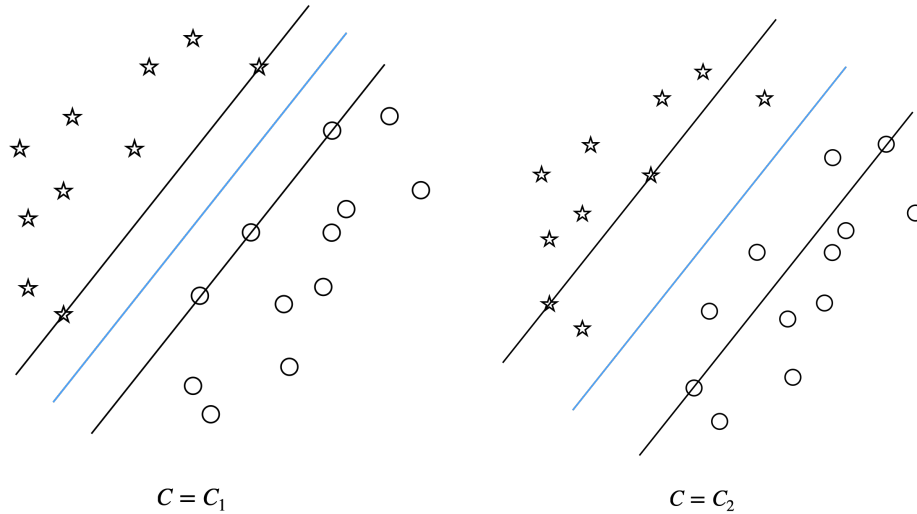
Consider the following Markov Decision Process (MDP) with discount factor $\gamma = 0.5$. Upper case letters A, B, C represent states, arcs represent state transitions; lower case letters x, y, z represent actions; signed integers represent rewards; and fractions represent transition probabilities. Discount factor γ is 0.5. Consider the policy π that takes action x in any state (i.e $\pi_i(s) = x$ for all $s \in \{A, B, C\}$). Compute the value function of this policy (i.e compute $V^\pi(A)$, $V^\pi(B)$, $V^\pi(C)$).

3 Support Vector Machines

You are given a dataset $\{x_i, y_i\}_{i=1}^n$ of n points for a binary classification task. Here, x_i 's are Euclidean vectors and $y_i \in \{0, 1\}$. You wish to use a soft-margin SVM to classify the points. Recall that the primal problem can be written as follows:

$$\begin{aligned} \min_{w, b, \xi} \quad & \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t. } \quad & \forall i, y_i(\langle w, x_i \rangle + b) \geq 1 - \xi_i \text{ and } \xi_i \geq 0 \end{aligned} \quad (1)$$

1. Generally speaking, what is the *main reason* for solving the dual of the optimization problem?
2. Let $\{\alpha_i\}_i$ denote the dual variables corresponding to the $y_i(\langle w, x_i \rangle + b) \geq 1 - \xi_i$ constraints. Suppose we solved the dual problem with $C = 5$ using a dataset of $n = 10$ points and found that the optimal α_i values were $[5, 4.99, 1.01, 1, 0.99, 0.01, 0, 0, 0, 0]$. List all the support vectors. You may list the indices of the points which correspond to the support vectors.
3. The two figures below show the decision boundary and the margin of an SVM trained on some labeled data with regularization parameters $C = C_1$ (left) and $C = C_2$ (right). The blue line represents the decision boundary, and the black lines from either side represent the size of the margin from the decision boundary. We have indicated the two classes using circles ($\circ$) and stars ($\star$).



- (a) Circle the points corresponding to the support vectors in *both* figures.
- (b) Of the following five statements about C_1 and C_2 , which *one* is the most accurate based on the figures above. Circle the right answer.

(i) $C_1 < C_2$
(ii) $C_1 \leq C_2$
(iii) $C_1 = C_2$
(iv) $C_1 \geq C_2$
(v) $C_1 > C_2$